

LC 2101: Detonate Maximum Bombs

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Target: Uber, Amazon

Problem Intuition & Notes

Input: $bombs[i] = [x_i, y_i, r_i]$

One bomb can detonate another if it is in range. **Chain reactions are allowed.**

Condition for Detonation:

If distance between A and B is less than $rad\ A$, then:

$$A \rightarrow B$$

This forms a *directed graph*. A can reach B, but B might not reach A.

Chain Reaction Example:

If $dist(A, B) < rad\ A$ and $dist(B, C) < rad\ B$, then:

$$A \rightarrow B \rightarrow C$$

Algorithm Steps

Step 1 (S1): Build the Graph

- For every pair (i, j) check if bomb i can reach j .
- Use the distance formula: $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \leq r$
- Or simply compare squared distances to avoid floating-point precision issues:

$$dx^2 + dy^2 \leq r^2$$

- If yes, then `adj[i].push_back(j)`.

Step 2 (S2): Depth First Search (DFS)

- Once the graph is built, start DFS from every bomb.
- We need to find the bomb that triggers the maximum number of detonations.
- Track the max count of detonated bombs across all possible starting points.

C++ Source Code

```

class Solution {
public:

    void dfs(vector<vector<int>>& adj, vector<int>& seen, int& cnt, int curr){

        seen[curr]=1;
        cnt++;

        for(auto &it:adj[curr]){
            if(!seen[it]){
                dfs(adj,seen,cnt,it);
            }
        }
    }

    int maximumDetonation(vector<vector<int>>& bombs) {
        int n=bombs.size();

        //har ek pair try krenge bomb ka
        //agar kisi do bomb ke beech ka distance radius se kam hua to graph
connect
        //eg A-B < rad(A) , A phatega to B phatega so A->B

        vector<vector<int>> adj(n);
        for(int i=0; i<n; i++){
            int x1=bombs[i][0];
            int y1=bombs[i][1];
            int r=bombs[i][2];

            for(int j=0; j<n; j++){

                if(i==j) continue;
                int x2=bombs[j][0];
                int y2=bombs[j][1];

                long long dx=1LL*(x2-x1);
                long long dy=1LL*(y2-y1);
                long long rr=1LL*r;

                if(((dx*dx)+(dy*dy))<=rr*rr){
                    adj[i].push_back(j);
                }
            }
        }

        //ek baar adj list ban gayi to ab har bomb ko detonate krke dekhenge ki
        //kis case me zyada se zyada bam phate

        int ans=0;
        for(int i=0; i<n; i++){
            int cnt=0;
            vector<int> seen(n,0);

```

```
        dfs(adj,seen,cnt,i);

        ans=max(ans,cnt);
    }

    return ans;
}

};
```

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